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The Nervous System

The Nervous system is made up of neurons who communicate via neurotransmitters.

This communication network takes in info from the world & the body's tissues, makes decisions, and sends back info & orders to the body's tissues.

Brain + Spinal cord = Central Nervous System.

The Peripheral Nervous System is responsible for gathering info & transmitting CNS decisions to other body parts.

Nerves are electrical cables formed from bundles of axons.
Nerves link CNS with sensory receptors, muscles & glands.

Ex. The Optic Nerve bundles 1 mil axons into a cable carrying messages from the eye to the brain.

Sensory Neurons carry messages from tissues and sensory receptors inward to the brain & spinal cord for processing; afferent

Motor neurons are efferent, carrying instruction from the CNS outward to muscles & glands

Between sensory & motor neurons, info gets processed by Interneurons, where our complexity lies.

We have few mil sensory neurons, few million motor neurons & billions of Interneurons.

PeNS { Somatic Nervous System
Autonomic Nervous System

The SoNS enables voluntary control of skeletal muscles. Ex. When smne taps ur shoulder, ur SNS reports current skeletal muscle state to your brain, triggering your head to turn.

ANS controls glands & internal organ muscles, Autonomic means 'self regulating'

ANS influences glandular activity, heart beat & digestion. One may consciously override this system, but it usually operates autonomously.

The autonomic nervous system has 2 subdivisions:

- The Sympathetic Nervous System
- The Parasympathetic Nervous System

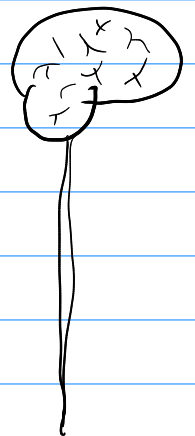
The SyNS arouses & expends energy
fight or flight

Accelerates heart beat, raises blood pressure,
slows digestion, raises blood sugar, cools you
with sweat, making you alert & ready for action.

When stress is over, the PsNS will
produce the opposite effect, conserving energy as it
calms you.

SyNS & PsNS work together to keep body
stable in homeostasis.

The complexity of the CNS emerges from the
process of neurons "talking" to each other.



Our brain enables our humanity:
Our thinking, feeling & acting.

By one estimate, our brain contains 128 bil neurons.

The brain's individual neurons cluster into work
groups called neural networks.

Neurons network with nearby neurons with which they can have short, fast, connections.

Each layer's cells connect with various cells in the neural network's next layer.

Learning occurs as experience strengthens connections

"neurons that fire together, wire together"

The Spinal Cord is a two-way information highway connecting the brain & PNS.

Ascending neural fibers send up sensory info & descending fibers send back motor-control info.

The neural pathways governing our reflexes, our automatic response to stimuli, illustrate the spinal cord's work.

A simple spinal reflex pathway, the reflex arc, is composed of a single sensory neuron and motor neuron. These often communicate through a spinal cord interneuron. Ex. the knee jerk reflex involves a reflex arc from the PNS to the CNS's spinal cord & then back out through the PNS.

The Pain reflex is enabled by a neural circuit. When ur finger touches a flame, neural activity is excited by the heat & travels via sensory neurons to interneurons in your spinal cord. These interneurons respond by activating motor neurons leading to the muscles in your arm.

The simple pain reflex pathway runs through the spinal cord & right back out, so your hand jerks away from the candle's flame before your brain receives the info.

If the top of your spinal cord was severed, you would still have reflexes, but would lose all voluntary control over your muscles and lose all sensation incl. pain & pleasure.